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SCHOOL OF HUMANITIES, LANGUAGES AND SOCIAL SCIENCE

DEVELOPMENT OF ENGLISH-BULGARIAN COMPREHENSION ASSISTANT

BY

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PRESENTED IN PARTIAL FULFILMENT OF THE FOLLOWING AWARD

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PREFACE

This paper discusses the architecture and the features of the English-Bulgarian Comprehension Assistant “Smarty”.

The project aims to test interface tools and advanced dictionary look-up techniques for part-of-speech tagging, multi-word expressions recognition and word sense disambiguation, which narrow the set of suggested translations and make comprehension of text in English faster and easier for non-expert readers.

Since there is no evidence for the existence of equivalent or comparable developments of comprehension assistant tools for English to Bulgarian or for any other language to Bulgarian, “Smarty” also aims to be the most sophisticated Bulgarian application for bilingual dictionary look-up.

The paper is divided in five chapters. The first one introduces the term "comprehension assistant" and discusses the need for intelligent dictionaries, supported by natural language processing technology. The second chapter presents previous developments and achievements in the field. The third one is devoted to the system we have developed and suggests a number of future improvements. The fourth chapter evaluates the results. Chapter five concludes the paper.

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1. Introduction to Comprehension Assistants

Still the most widely used software tools, created to help comprehension are the conventional computer dictionaries, which ironically work the simplest possible way.

This chapter emphasizes the inadequacy of this situation to the state-of-the-art of Natural Language Processing technologies and gives suggestions about how a NLP-supported dictionary should look like.

1.1. Conventional Computer Dictionaries

Globalization and explosion of Internet, almost exclusively in English at the time, made computer "To English" and "From English" dictionaries a must-have utility in late 90-ies and early 2000s in many non-English-speaking countries, including Bulgaria. Although the quantity and variety of information, available in other languages grew immensely, yet English is, and is supposed to be the ultimate way to spread messages around the world to as many people as possible.

Since acceptable open-domain machine translation systems do not exist and what is available can not satisfy the need of proper and accurate translation, the masters of mass quasi-translation market became a lot of off-line and on-line multilingual computer dictionaries.

However, while actually being much more useful and handy than the classic paper dictionaries, almost all available computer dictionaries provide roughly the same limited set of functionality and suffer from the same simplicity. The user just enters a word and receives an article from scanned or typed paper dictionary.

Few programming enhancements - predictive typing and automatic translation of a word, copied by the user to the clipboard of the operating system - quickly became a sort of standard, but yet these simple programming tricks do not save the tedious job of browsing for the appropriate sense, when the queried word has many of them.

1.2. Conventional English-Bulgarian Computer Dictionaries

The most popular English-Bulgarian dictionaries - **SADictionary** and **AEnglish** - differ very little in their interfaces and share almost the same functionality and lexical database.

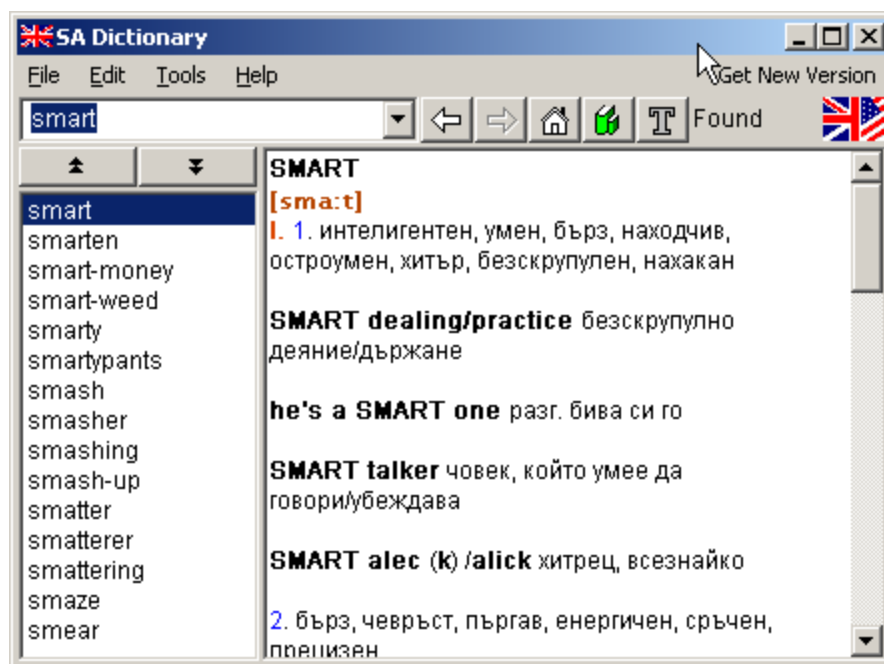


Fig.1 SA Dictionary

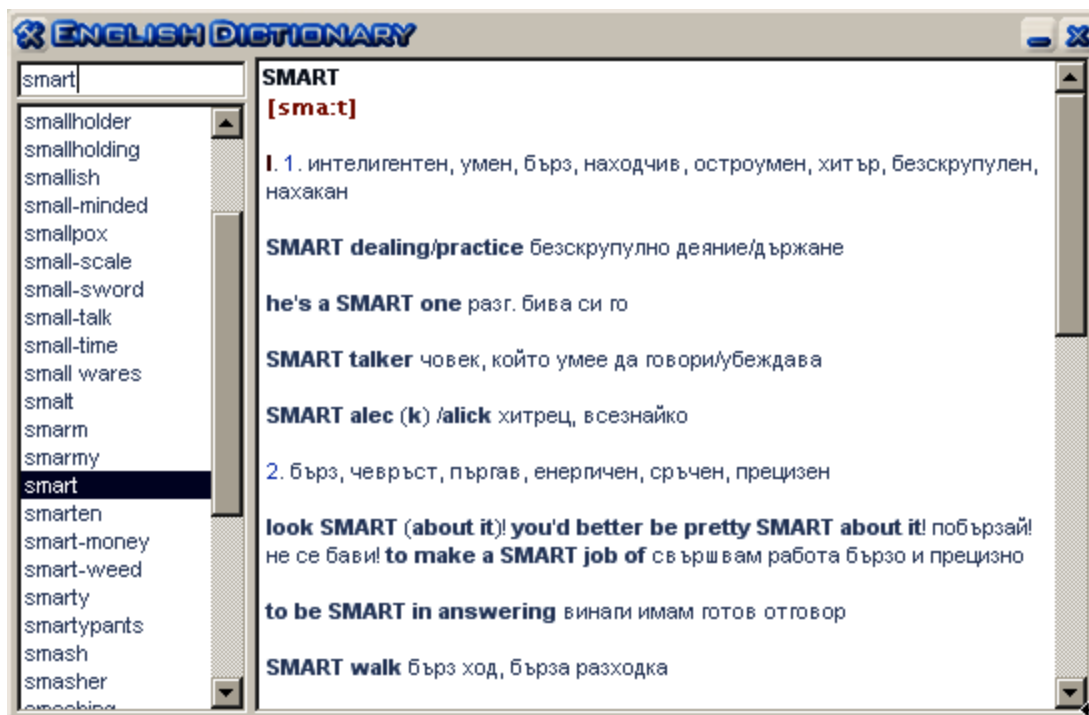


Fig.2 AEnglish Dictionary

Even though they both have examples for multi-word expressions (MWE) in their dictionary entries and they do display them neatly in bold font, these dictionaries do not search for matches, so if more than one word is entered, in the best case the user will get an article for the word which begins like the first word from the input.

Using SA Dictionary, the only way for recognition of MWE is scrolling the article and search for the expression by a naked-eye, assisted solely with bold font and capitalized keywords. Alphabetical order is unavailable.

AEnglish has "Find" dialog-box for the entries, but it would capture only exact matches.

Obviously, even basic NLP-processing is missing.

1.3. Limits of Conventional Dictionaries

While it have to be admitted, that conventional bilingual computer dictionaries work fine in a lot of occasions - dealing with unambiguous words with single or very few senses, studying new words - the usefulness of this simple technology decreases with the increase of the length of the list of senses.

Many English words are extremely ambiguous at two levels: they could play different roles as part-of-speech – this is **PoS-ambiguity** - and in each of these roles they may have a lot of sub-meanings – this is **lexical polysemy**, – or they may have one or several **homographs** – words which share the same string of letters, but have completely different meanings.

For example, **WordReference.com** suggests about **85 translations** of the word "run" in French - as a verb and as a noun.

If a dictionary with the same database has the information that "run" is, e.g. a **noun** and not a verb, that would reduce the list of translations to about **25** (the numbers are similar for English to Bulgarian, referring to SA Dictionary). Although still not a precise suggestion, the progress is very good, and it is important to note that this is not a difficult problem to solve, since today's part-of-speech tagging is a robust technology with very high precision, reaching over 95-97% for some languages.[7, Ch. 11]

If the word is used with a preposition or next to particular collocating word, the dictionary may try to recognize corresponding compound form or a multi-word expression. In case of success, this step could reduce the senses to **one or very few**.

Going further, current NLP technology allows Word Sense Disambiguation (WSD), based on analysis of the context of the given word, for solving lexical polysemy. State-of-the art precision of WSD is up to 75% [7, Ch. 13].

1.4. Comprehension Assistants

"Comprehension assistant" and "Intelligent dictionary" are synonyms of:

Advanced multilingual computer dictionary, which is not merely an electronic copy of a paper dictionary, but is enhanced with variety of natural language processing techniques for narrowing the set of suggested translations.

While standard dictionaries are queried with **single words**, unrelated to any others, comprehension assistants process *words in context*.

While standard dictionaries always display all the information, linked to a given **word**, comprehension assistants aim to give the *most relevant information for the particular usage of a lemma*.

2. Previous Work

The usefulness of comprehension assistants is undoubted and proved by systems initiated more than 10 years by Xerox LocoLex.

Unfortunately this type of tools still lacks popularity among regular users, due to its high-cost or unavailability.

2.1. LocoLex

Rank Xerox has developed the first comprehension assistant at PARC, XRCE between March 1994 and March 1996 within the framework of the COMPASS European project. *“LocoLex uses part-of-speech tagging and idiom recognition to provide users with the translation relevant to the word in its particular context, facilitating and accelerating the process of comprehension”*. [8]



Fig 3. LocoLex

2.1.1. LocoLex Features

- User points a word from a text in a window of LocoLex.
- The context around the word is split into words.
- Each word from the sentence is normalized to a standard form for unified dictionary look-up.
- Words are morphologically analyzed and possible parts-of-speech cases are generated.
- The most likely syntactic usage of the pointed word is disambiguated, basing on surrounding words.
- The relevant entries are found, including possible homographs and compound forms.
- Irrelevant sections are eliminated, using morphological analysis and disambiguation phases.
- Special or idiomatic usage is identified.
- Only the most appropriate translation is displayed, based on the part of speech and surrounding context. [7, Ch. 38]

These features became the minimum standard for an application to be accepted as a “comprehension assistant”.

2.1.2. Evaluation

H. Feldweg and E. Breidt reported "overwhelmingly positive" results even at the first test phase.[8] A second test phase, performed with an improved version of the prototype of Locolex, had been supposed to be conducted in the beginning of 1996 [8]. However, the final paper and evaluation of the first Locolex was not found.

2.1.3. LocoLex with Word Sense Disambiguation

A rule-driven semantic word sense disambiguation was developed as an extension of Locolex in late 90-ies, initially only monolingual - for English, - even though the system extracts its disambiguation rules from the same Oxford-Hachette English-French and French-English dictionary, used in LocoLex. [8]

A bilingual Word sense disambiguation extension of LocoLex was created, as well - unsupervised rule-based semantic tagger, which works on all input words. Semantic disambiguation rules are directly extracted from dictionary examples and their sense numberings. [7, Ch. 38].

2.2. Morphologic MobiDic, MetaMorpho and MobiCat

2.2.1. Goals

These applications emphasize the complexity of modern graphical user interface applications, which are loaded with too many windows and menus. They aim to provide minimalistic user-friendly interface for their comprehension assistants, allowing their systems to be easily and transparently integrated in third-party applications.[12]

2.2.2. MobiDic and MobiMouse

Morphologic's ideas about interface are illustrated in MobiDic, MobiMouse and MobiMouse Plus series. Their software is capable to detect and translate any word on the screen, including menu commands, and the result is displayed in the corner of the screen - user can always look if he or she feels uncertain, yet without being distracted if he or she does not need comprehension assistance at a given moment.

2.2.3. MobiCat with MetaMorpho

MetaMorpho is a powerful sentence-level English-Hungarian machine translation system, based on Pattern Based Machine Translation - a hybrid method, combining probabilistic Example Based Machine Translation and a deterministic Rule Based Machine Translation. [12]

MobiCat is a commercially available application of MetaMorpho.

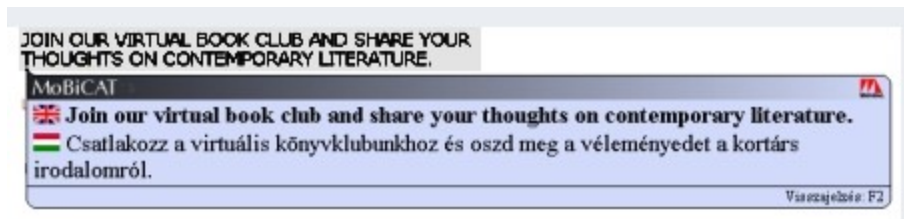


Fig 4. MobiCat

3. Smarty - English-Bulgarian Comprehension Assistant

This chapter presents the design and the architecture of “Smarty” and suggests possible improvements for future work.

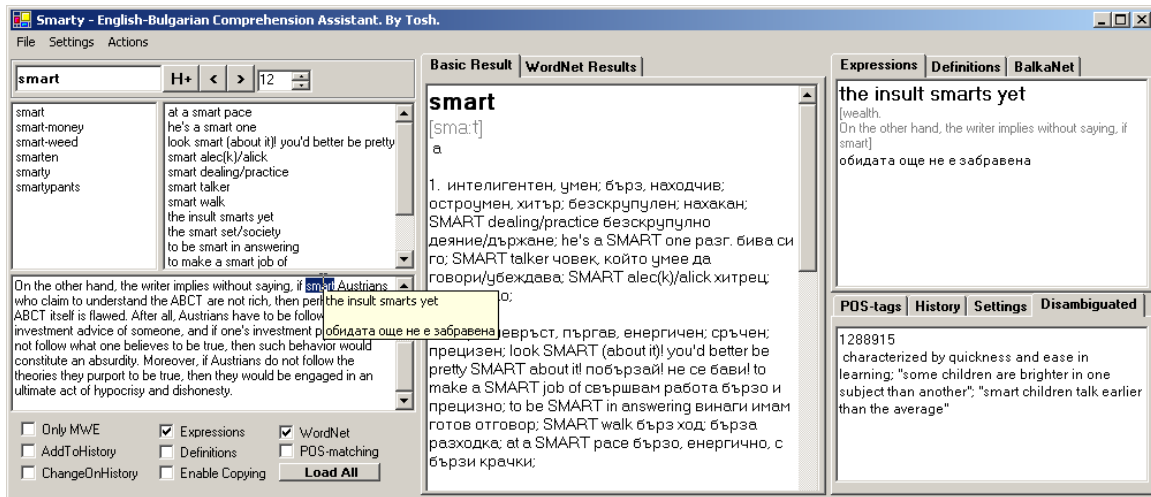


Fig 5. Smarty in action

3.1. Linguistic Resources

“Smarty” is based on three linguistic resources:

1. English-Bulgarian Standard Dictionary.
2. Princeton University WordNet 2.0.
3. BalkaNet – WordNet for Bulgarian.

3.1.1. English-Bulgarian Standard Dictionary

Initially this was a text-file, named “en.txt”, distributed freely on the Internet as a base for development of conventional dictionaries. It consists of about 52000 dictionary entries, each of them containing a word; a transcription; and an entry with part of speech information, a list of senses and, for some of the words, examples of multi-word expressions or usage.

The main drawback of the database is the “condensation” of information, which complicates the parser for expanding dictionary entries into a more convenient format.

For instance, the boundary between an example of a multi-word expression or a usage, and the corresponding translation is only the character set used – Latin versus Cyrillic. Probably, due to scanning and optical character recognition, there are mistaken letters, which look the same in Cyrillic and Latin – e.g. **е, н, о**. This ambiguity must be corrected by heuristics; otherwise the expressions are incorrectly extracted. In other cases, some Cyrillic letters “З” are placed instead of the number “3”.

Another complication are English words in brackets, inserted between the words in the definition in Bulgarian.

In addition, entries for words with many senses, like “run” or “go”, have references from one sense to another sense of the same word, by expressions like “run for 1)= run1 17.”

3.1.2. WordNet 2.0

Princeton University’s WordNet is a large lexical database, a collection of synonym sets of words - “synsets” – structured by part-of-speech and numerous types of semantic relations. Due to the richness of the structural information in WordNet, it is used as a resource and tool for various NLP and Computational Linguistics tasks, such as word sense disambiguation, machine translation, automatic generation of multiple-choice tests and many others. [7, 11]

Although current latest version is 3.0, “Smarty” used version 2.0 from 2003, due to its alignment with the third important resource for our research – BalkaNet (see below). WordNet 2.0 consists of over 115000 synsets, and the total count of all unique lexemes of noun, verbs, adjectives and adverbs is 144309 with 203145 word-sense pairs in total. [10]

Our system relies on WordNet in three directions:

1. Glosses with various senses of the words in English are used to back-up the basic dictionary in case the word does not have an entry there. The gloss is not a translation, but may help dictionary user to grasp the meaning, because it is supposed that he or she has some knowledge of English.
2. Glosses in Bulgarian from BalkaNet are presented when applicable, by mapping to WordNet.
3. **Word Sense Disambiguation is performed, using the glosses in WordNet to reduce the offered list of senses, ultimately to one.**

3.1.2.1. Structure of WordNet 2.0

The core of the database is organized in four index and four data files for **nouns**; **adjectives** and **adjective satellites**; **adverbs** and **verbs**.

Index files list the synsets to which a given lexeme belongs.

Data files contain the **lemmas – individual words or collocations** – of the synsets; lists of semantic and lexical relations between different synsets - hyperonymy, hyponymy, synonymy, antonymy, holonymy, meronymy etc. – and short gloss and/or examples of usage for the lemmas.

The lemmas of the synsets and their corresponding glosses are our resource for performing word sense disambiguation (see below).



3.1.3. BalkaNet

“The Balkan WordNet aims at the development of a multilingual lexical database comprising of individual WordNets for the Balkan languages. The most ambitious feature of the BalkaNet is its attempt to represent semantic relations between words in each Balkan language and link them together in order to develop an on line multilingual semantic network.” [2]

BalkaNet was developed for Bulgarian, Czech, Greek, Romanian, Serbian and was aligned to WordNet 2.0, i.e. the synsets of any pair of these lexical semantic databases could be linked directly.

	Bulgarian	Czech	Greek	Romanian	Turkish	Serbian
Synsets	21441	28456	18461	19839	14626	8059
Nouns	14174	21009	14426	13345	11059	5919
Verbs	4169	5155	3402	4808	2725	1803
Adjectives	3088	2128	617	852	802	324
Adverbs	9	164	16	834	40	13
Literals	44956	43918	24366	33690	20310	13295
Literal/Synset	2.1	1.54	1.33	1.7	1.39	1.65
BC1s	1218	1218	1218	1218	1220	1219
BC2s	3471	3471	3462	3471	3479	3469
BC3s	3827	3827	3825	3827	3794	1369
Domain Specific Synsets	2065	304	238	286	300	305
Balkan Specific Synsets	220	257	309	151	103	117
Language Specific Synsets	116	257	52	545	204	206
Language Internal Relations	28599	25683	24368	25885	19834	12787

Table 1. Statistics for the final version of BalkaNet. [2]

“Smarty” uses a smaller version of Bulgarian part of BalkaNet, consisting of about 15000 synsets.

3.2. Architecture of Smarty

The System can be divided in 11 major components:

1. Interface.
2. Wrapper of SharpNLP PoS-tagging library.
3. Additional Tokenizer and Token-With-Context Extractor.
4. Basic bilingual dictionary database.
5. Predictive Typing Processor.
6. Standard Dictionary Look-up.
7. Dictionary Entries Analyzer.
8. Normalizer (Lemmatizer and Forms Generator).
9. Multi-Word Expression Matcher.
10. WordNet 2.0 and BalkaNet Wrapper and Mapper.
11. Word Sense Disambiguator.

3.2.1. Interface

The limited time for development of just three months forced us to emphasize much more on the natural language processing aspects than on the graphical user interface (GUI). For that reason some of the windows were left in an unpolished stage; yet we had to spend some time for research and development of the sophisticated GUI, which is an important part of the comprehension assistance and query speed-up, in comparison to conventional computer dictionaries.

3.2.1.1. Main Input and Output in Standard Mode

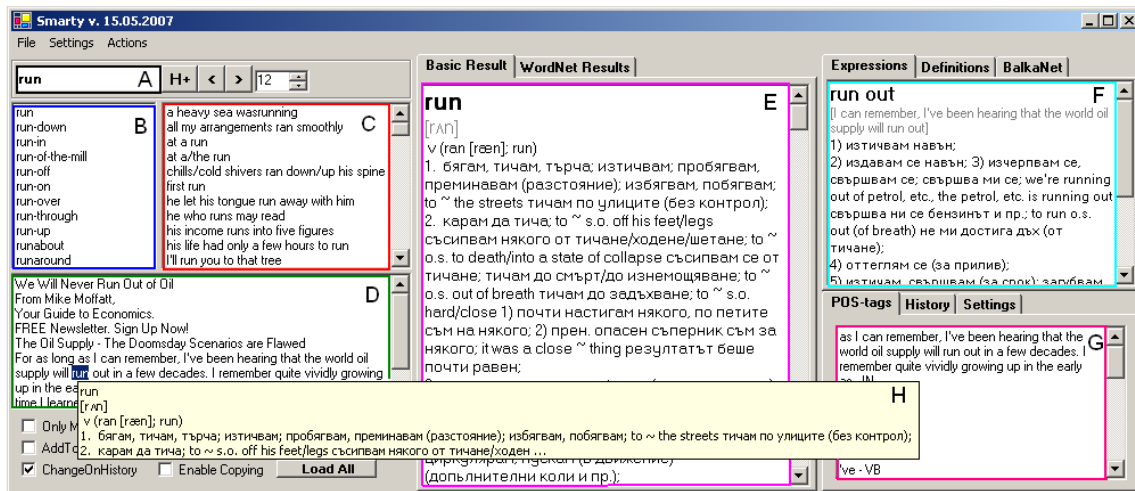


Fig 6. Input and output boxes in standard mode

- A. Query Textbox
- B. Suggested (Predicted) queries
- C. Alphabetically sorted known multi-word expressions
- D. Free text input
- E. Basic dictionary output
- F. Suggested multi-word expression
- G. Auxiliary PoS-tagged output
- H. Optional definition/expression tool tip

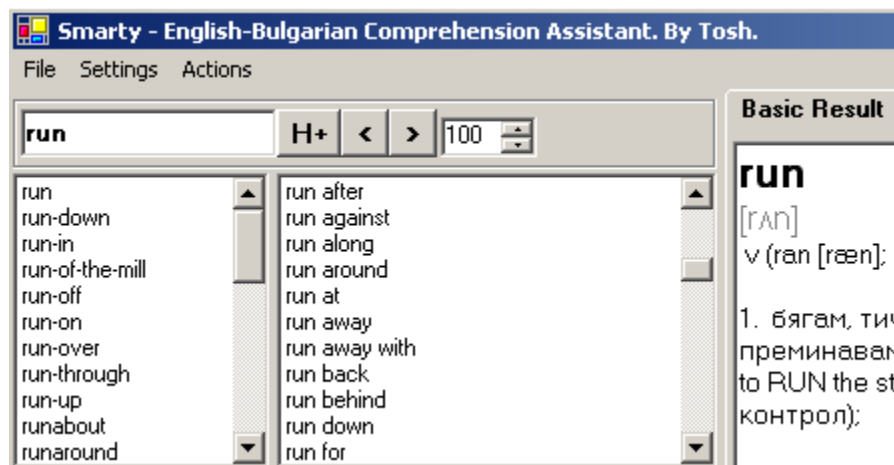


Fig 6. Predicted queries and the list with multi-word expressions

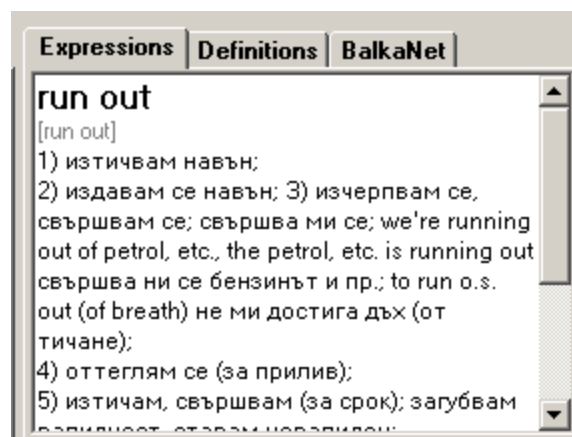


Fig 7. Multi-word expression translation

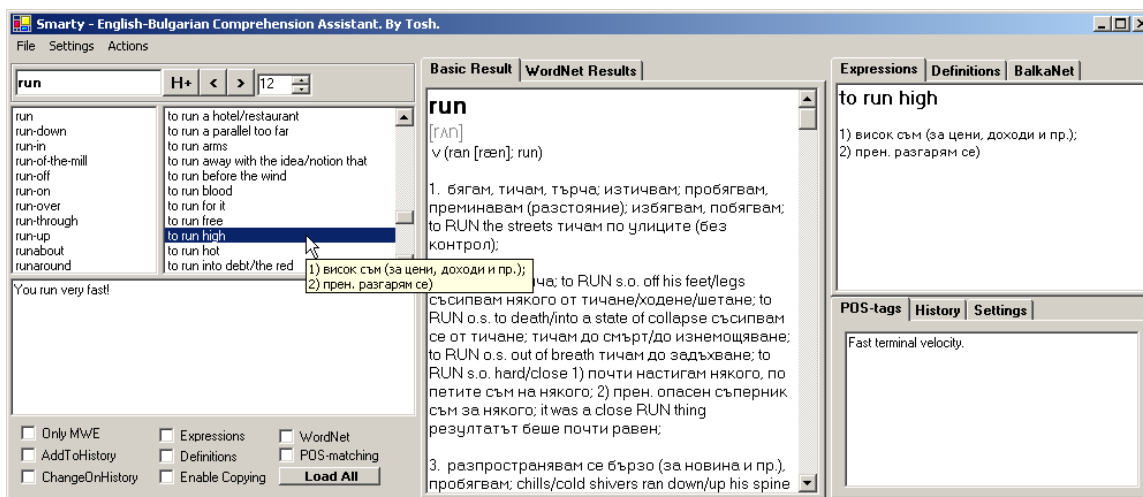


Fig 8. Multi-word expression tooltip

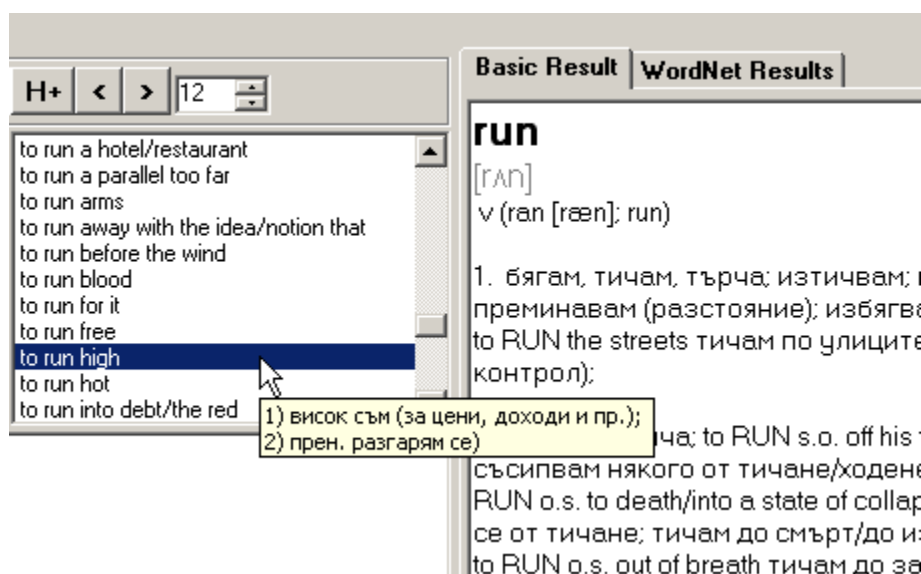


Fig 9. Multi-word expression tooltip – close-up view

3.2.1.2. WordNet/BalkaNet Mode with Word Sense Disambiguation

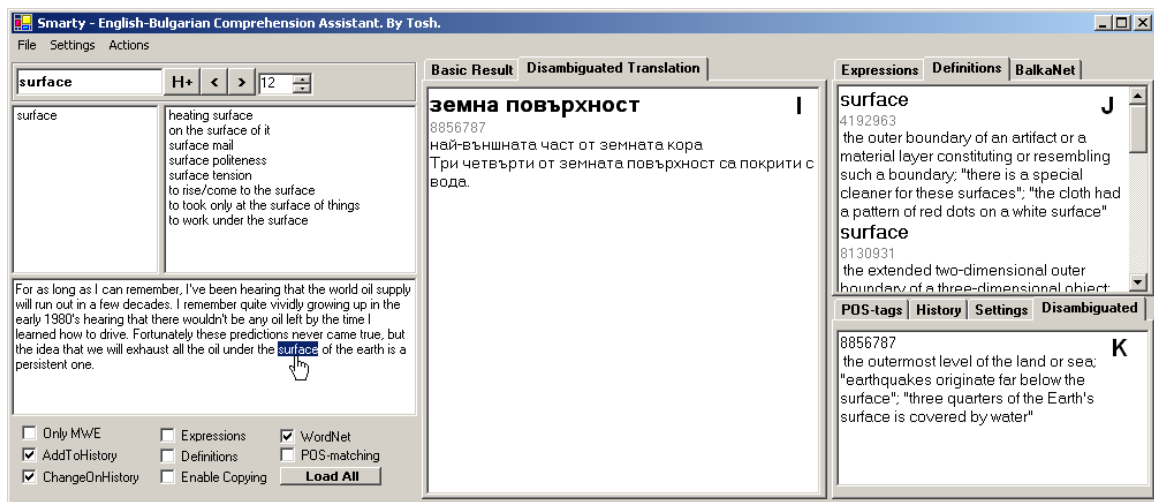


Fig 10a. Output windows boxes in Word Sense Disambiguation mode

- I. Disambiguated translation
- J. Full list of possible translations from BalkaNet
- K. Disambiguated sense in English from WordNet.

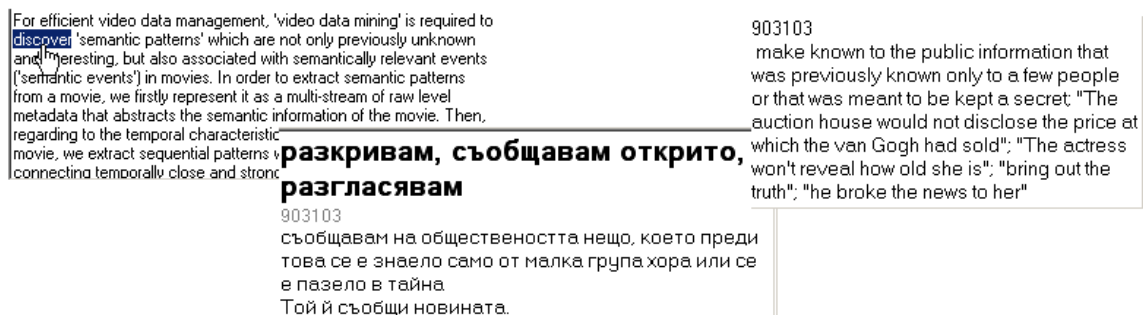


Fig 10b. Word Sense Disambiguation windows

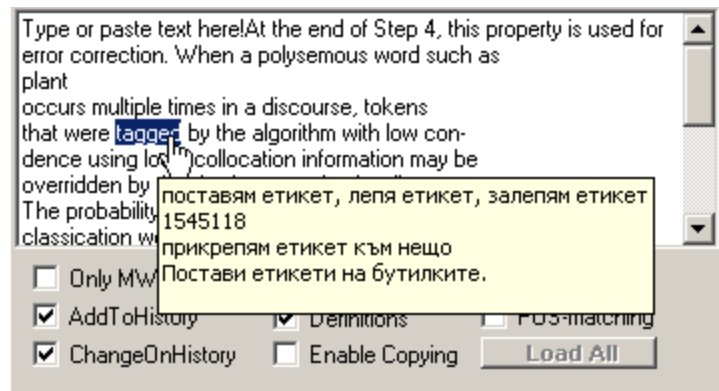


Fig 10c – Disambiguated translation may appear also as a tooltip, on a right mouse click

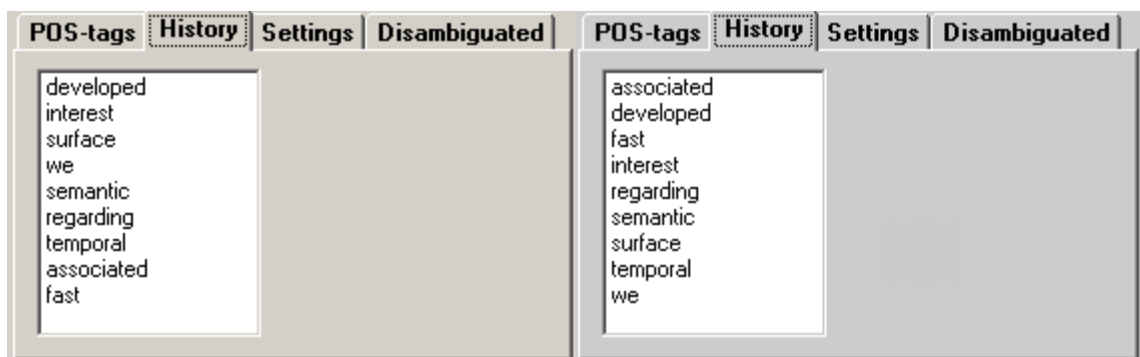


Fig 11. History of queries could be sorted by time and alphabetically

L. History of queries

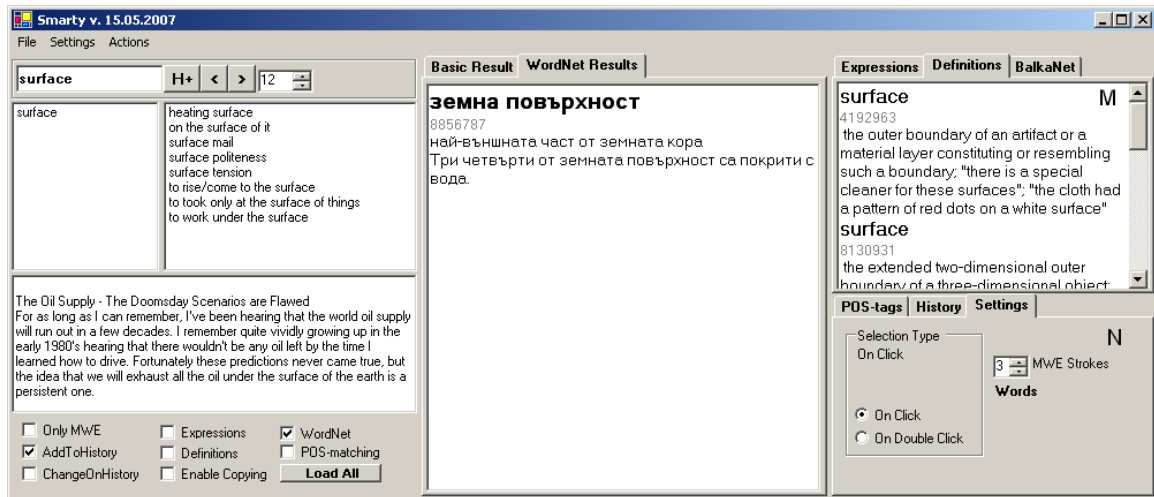


Fig 12.

M. Full list of senses from WordNet

N. Auxiliary options panel

3.2.1.3. Other Controls and Options

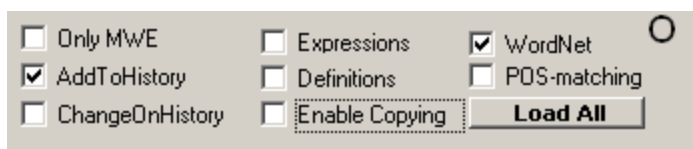
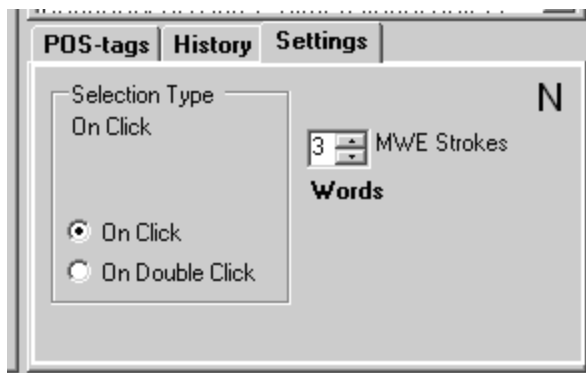


Fig 13. Options

N. Auxiliary options panel

O. Main options

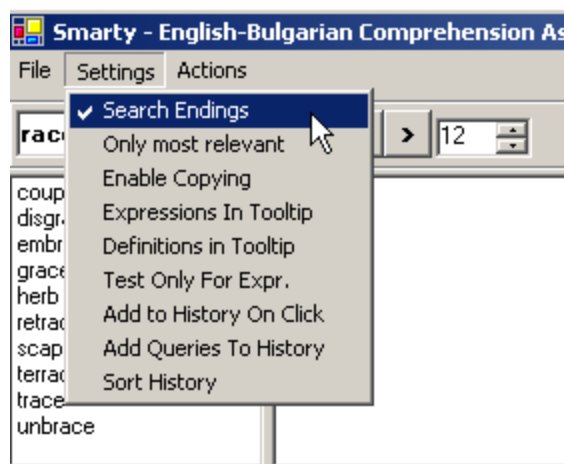


Fig 14. Suffix search

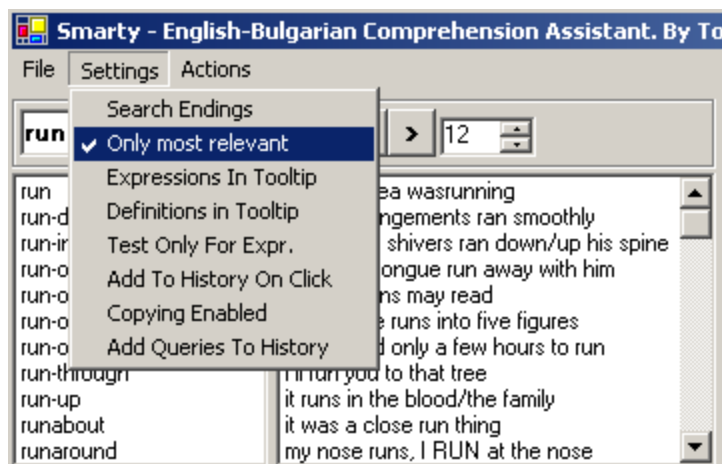


Fig 15. Display only the entries with matching PoS-tag

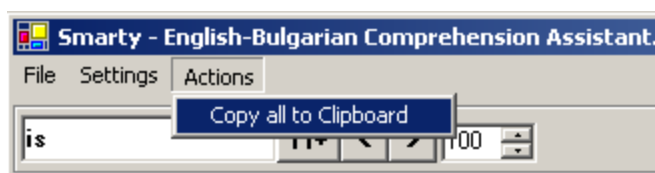


Fig 16. Copy All to Clipboard – the text from all dictionary windows is copied



Fig 17. Query Panel

Q. Query panel



- Query Textbox



- Add to history list



- Move a step back/ahead in history list



- Maximum number of predicted query suggestions

3.2.2. The Interface in Details

3.2.2.1. Basic Input

The basic mode of operation of “Smarty” is similar to that of conventional dictionaries: the user types or pastes a word or a short expression in the query textbox.

In case of typing, the program tries to answer user’s query as fast as possible: on each keystroke (or after text is pasted):

1. A search through the words database is performed and a list of words **beginning like the query** is shown.
2. Suggestions about possible **continuations** (default mode) or possible **beginnings** (suffix mode – menu P) are displayed.

3. Without additional user actions, **appropriate dictionary entries, matching the query, are immediately displayed**. If the entered query does not match an entry from the list with suggestions, the program “makes a guess” and displays the one which is most similar.
4. Once a few keystrokes – the number is specified in N – or when Enter is pressed, **the query is matched to the list of multi-word expressions. This is useful for checking compound forms like “run away”, “go down” etc.** The suggested expression – either the correct one or just an example - is displayed in window F.

Multi-word expression matches could be based **on the lexemes** or on the **part-of-speech tags** – that is specified in N and O by clicking on “**Words**” or by checking “**PoS-matching**” check-box.

5. If the user presses Enter, the PoS of the word is determined and the dictionary word is **lemmatized**; the dictionary takes into account the original form of the input, too. Note that, a single word in English, without a context, can represent multiple PoS.

In case entries for both lemmatized and original form of the query are found, the dictionary displays first, on top, the basic output of the entry in its original form, even in cases when the PoS-tag doesn't match, presuming that this entry could be the closest to the query, since the PoS-tagger is not 100% accurate.

The lemmatized and PoS-tagged word is also checked in WordNet database. If the word exists in a synonym set, a list of all relevant definitions is displayed in M; after that, the program aligns the results with BalkaNet and if corresponding synsets are found, all possible definitions in Bulgarian are also shown in windows J and I.

Even in this basic mode, the intelligent dictionary proves to be a more powerful tool than its conventional English-Bulgarian dictionary rivals, thanks to the *Multi-Word Expressions matching, WordNet and BalkaNet look-up and suffix-searching*.

These functions will be discussed below.

3.2.2.2. Free Text Input

The real advantages of “Smarty” over simple dictionaries emerge as soon as the user pastes or inputs complete text in the Free Text Input window. In this mode the context around the word could be effectively used to narrow down the number of possible translations.

In this mode, the sequence of actions is as follows:

1. User clicks on a word in any window with text, including windows with results from previous queries with either the left or the right mouse button.
2. Context and the word are tokenized and PoS-tagged.
3. The word is lemmatized.
4. The most relevant entry from the standard dictionary is decided like in point 5. for the basic input.
5. Query textbox is filled with the word and suggestions for other forms of the word are generated.

If the left mouse button is pressed and tooltip-options “**Expressions**” or “**Definitions**” are checked, a tooltip with the top lines of the definition or with suggested expression are shown near the mouse pointer.

If the right mouse button is pressed, then a word-sense disambiguated translation appear as a tooltip, after passing points 7. and 8.

If “**Only MWE**” is checked, the program does not translate clicked words, but only tries to match multi-word expressions.

“**On click**” or “**On double-click**” in the auxiliary options panel N select the way the dictionary is queried from the free text input.

“Enable copying” disables translation on click and allows the user to select and copy text from any of the output or input windows.

“Add to history” – adds clicked words to history list.

“Change on history” – when a word from the history list is clicked, the query textbox is filled with the pointed word and related forms, and additional queries are predicted.

6. Multi-word expression matching is performed and an expression is suggested.
7. Relevant definitions from WordNet and BalkaNet are extracted and displayed.
8. Word-sense disambiguation is performed on the context. A disambiguated gloss in English is shown and **a disambiguated translation is suggested in Disambiguated Translations window I or in a tooltip.**

3.2.2.3. Tooltips

Tooltips with the top part of a dictionary entry, or with suggested multi-word expression, pop-up after clicking the left mouse button. The right mouse button activates a tooltip, containing disambiguated translation in Bulgarian.

A tooltip window with translation of a multi-word expression pops-up after a click on the list with expressions, and the content of that tooltip window could be easily copied to the clipboard by pressing F3.

3.2.3. SharpNLP PoS-tagger

Part-of-speech tagging, or PoS-tagging is one of the crucial NLP-techniques, used in “Smarty”.

Pre-processing tokenization and then PoS-tagging are done every time a dictionary is queried with a context or with a single word.

Smarty uses GNU SharpNLP PoS-tagger. Technical details about its operation are not available, but it is considerably robust for a test system.

This is a list of the most important tags that SharpNLP produces:

NN, NNS – Noun in singular, Noun in plural.

VB - Verb in present tense.

VBZ - Verb in present in 3-rd person.

VBG – Ing-form of a verb.

VBD – Verb in past tense.

VCN – Verb participle.

JJ – Adjective

RB – Adverb

IN - Preposition

DT - Determiner

A subtle post-processing of the output is done, in cases of strings of attributive nouns. SharpNLP tags them as simple nouns, while “Smarty” converts the attributive nouns into adjectives.

Also, during the process of lemmatization, words that end in “-ing” and are tagged as nouns (NN), are regarded as verbs.

3.2.4. Token-With-Context Extractor

This component tokenizes the word that is pointed with the mouse in Free Text Input window or any other window with text, and extracts the context around it, using tokenizer. The tokenizer extracts whole words only, thus avoiding sending truncated word fragments to the PoS-tagger.

3.2.5. Predictive Typing Processor

Predictive typing processor is based on a tree structure, internally called “LettersNet”, which describes a deterministic finite-state automaton, representing all the words from the standard dictionary database.

Positions in the words (first letter, second letter, etc.) are states, and the letters are transitions between states.

Each node can be continued with none, with zero or with all of the 26 letters of the English alphabet and with some special characters – space, hyphen and apostrophe. Last positions of the words are marked as final states of the automaton.

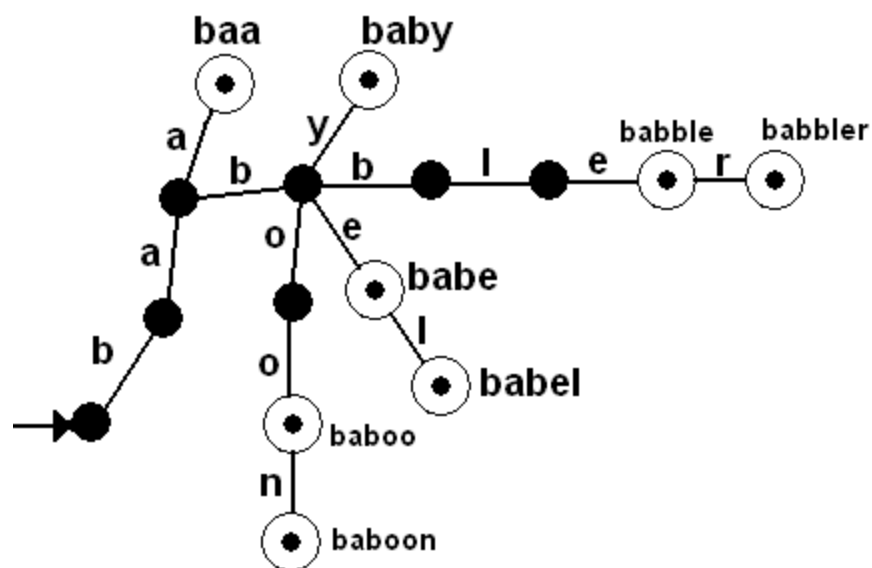


Fig 18. “LettersNet” finite automaton

This structure provides a linear complexity algorithm for search and almost “free” string sort, but at the cost of memory overhead.

Smarty uses two “LettersNet” structures – one for default mode, built from first to last letter of the words; and one reversed - i.e. the last letters of the words are fed first – which is used in suffix-mode.

3.2.5.1. Suffix Mode

This mode is useful for exploration of words from Latin, French etc. origin, and for writing poetry by performing rhyme search.

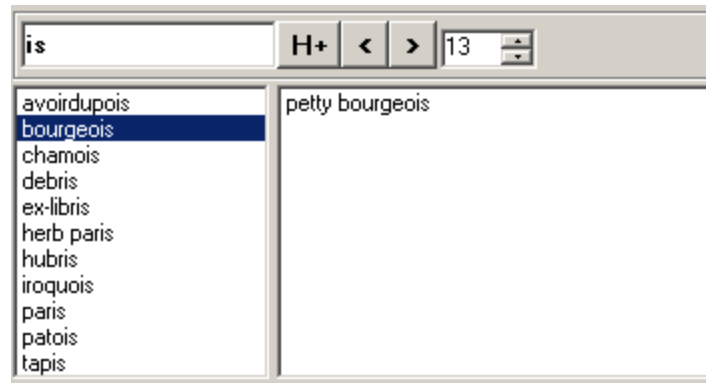
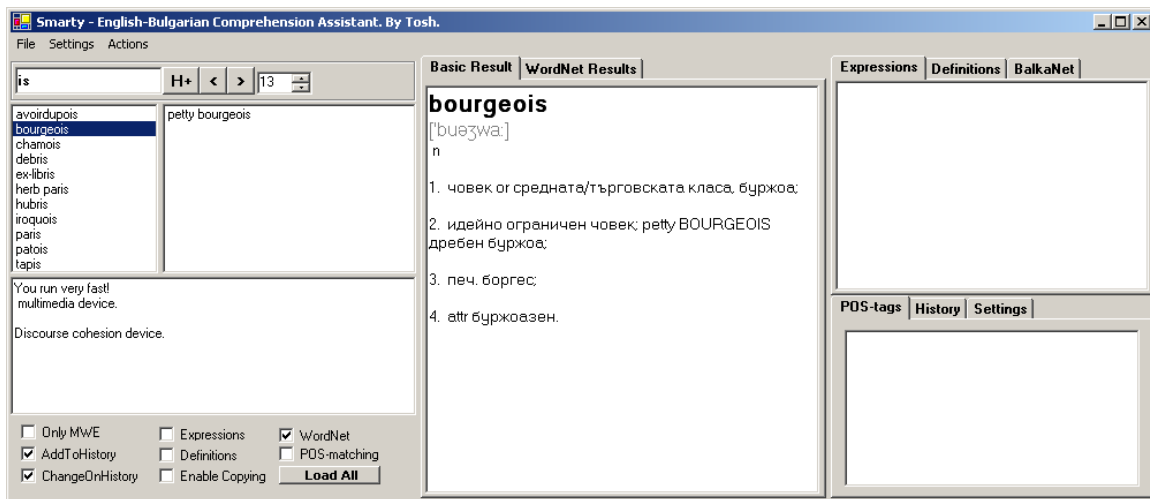


Fig 19. “Smarty” in suffix mode

To the best of our knowledge, Smarty is the first English-Bulgarian dictionary, capable of suffix-searching.

3.2.6. Dictionary Entries Analyzer

This component parses dictionary entries and extracts multi-word expressions on-the-fly, each time a word is queried.

The conventional dictionary database is an imperfect scan of a paper dictionary, intended to be read by humans; thus, a clear and explicit format definition was not available. For this reason, heuristic rules that explore the entries in a human-like manner were developed to deal with the ambiguity and errors. The “condensation” of the information, mentioned earlier, caused some difficulties (see also 3.1.1).

The result is a complicated scanner, which has to take into account open and closed brackets, numbering and abbreviations. The scanner works together with the Normalizer, presented below, which supplies the Analyzer with different forms of the words, in order to let it capture variations.

3.2.7. Normalizer

The role of the Normalizer is to find the lemma and, in some cases, to generate forms of a given PoS-tagged word .

Lemmatization part of the Normalizer is used for every query, and is activated either by typing or by clicking in any input window. Lemmatization is used also by Multi-Word Expression Matcher and by Word-Sense Disambiguator.

Forms Generator helps Dictionary Entries Analyzer to detect multiword expressions, because while it does not lemmatize the whole context, it scans it.

In Basic mode, the Forms Generator can expand a verb in the present tense into its past tense and participle forms.

3.2.7.1. Lemmatizer

Lemmatization of nouns and verbs in “Smarty” is based on simple morphological rules, a list with irregular verbs and irregular plural nouns, and the standard dictionary database.

For example, the rules for verb lemmatization are given below, where *Verb* is the input string.

0. If the Verb is any of (be, is, am, are, was, were)
 Then Return “be”
 Else
 If the Verb is “has”
 Then return “have”
 Else
 If the Verb is in any of (lay, lain, lied)
 Then Return “lie”
1. If the Verb is found in the list of irregulars – return the Infinitive.
 Else
2. If the Verb ends with “ed”
 3. If in the Dictionary exists a form which is same like the Verb without the last “d”.
 Then Return that form. //scribed - scribe
 Else
 4. If the Verb ends with double consonant before “ed” and it is not “l” //rolled - roll
 Then Return the Verb, shorten with 3 characters //rib – ribbed
 Else
 Return the Verb, shorten with 2 characters.
- Else
5. If the Verb ends with “ing”
 6. If the Dictionary has a form, which is like the Verb without the last “ing”, + “e”.
 Then Return the form from the Dictionary. //taking – take

Else

7. If the Verb ends with double consonant before “ing” and it is not “I” //calling - call

Then Return the Verb, shorten with four characters. //running – run

Else

8. Then Return the Verb, shorten with three characters. //flying - fly

Else

9. If the Verb ends with “ss”

Then Return the Verb

Else

10. If the Verb ends with “ies”

Then Return the Verb, shorten with three characters, and add “y” //flies - fly

Else

11. If the Verb ends with “es”

If the Dictionary has a form which is like the Verb without the last “s”

Then Return the form from the Dictionary. //takes – take

12. If the Verb ends with “s”

Then Return the Verb, shorten with one character.

Else

Return the Verb.

3.2.7.2. Forms Generator

The Forms Generator uses similar algorithms, but in reverse direction in order to construct a list of multiple forms. However, this sub-component was not fully developed, because only a generator of present-past-participle forms of verbs was needed.

3.2.8. Multi-Word Expressions Matcher

After Dictionary Entries Analyzer propagates a list of expressions, detected in the relevant active dictionary entry, Multi-Word Expressions Matcher is supposed to recognize an expression in the context of the pointed word.

Even if the Matcher is unable to recognize one particular expression with certainty – which is often the case – the Matcher suggests an expression which looks most similar to the context anyway. In the worst case, the example may just enrich user's knowledge.

Basic characteristics of the Matcher are:

- Two matching algorithms are selectable: **Words-Matching** and **PoS-Matching**.
- Multi-word expressions are stored in a format, regarding the “back” and “forward” content of the expression, where the main word is in the “centre”. E.g.:

“he let his tongue run away with him” is stored as:

Back: he let his tongue

Center: run

Forward: away with him

- Before performing matching, the tokens from the context are lemmatized in order to capture variations.
- The context that is to be matched is also stored not just as simple text, but with “back” and “forward” subsections for better accuracy. Appropriately sized sub windows of words from the context - back and ahead - are selected for each particular multi-word expression that is matched.
- Both algorithms rely not only on the absolute number of matched tokens, but also take into account the ratio of matched tokens to number of tokens, and the length of the matched expressions.

- Fully matched expressions have higher priority than partially matched, but fully matched longer expressions have higher priority than fully-matched shorter ones.
- In this version, “Smarty” always guesses only one expression, assumed to be the most similar to the context.

3.2.9. WordNet and BalkaNet Wrapper

This component indexes, queries and links together WordNet and BalkaNet lexical databases.

One of the scenarios of using the Wrapper is in standard mode, when no context is given. Then the Wrapper is queried with a “bare” token.

The Wrapper PoS-tags the token. If the token with the tagged part-of-speech exists, the Wrapper returns a list, consisting of a synonym set and glosses/examples-of-usage for each member of the synset.

Glosses are displayed in window M with their synset offset for reference.

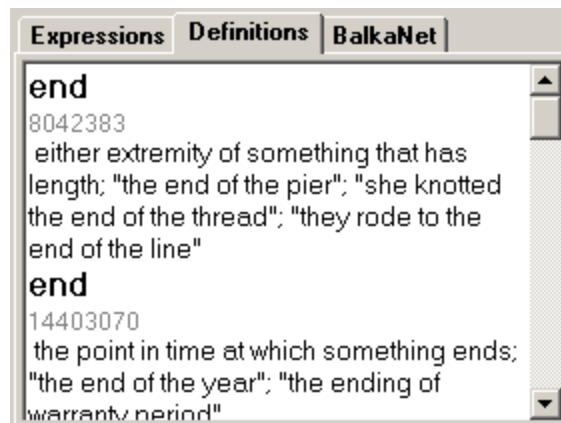


Fig 20. WordNet definitions

Then BalkaNet is checked for corresponding synsets; if found, they are displayed, also:

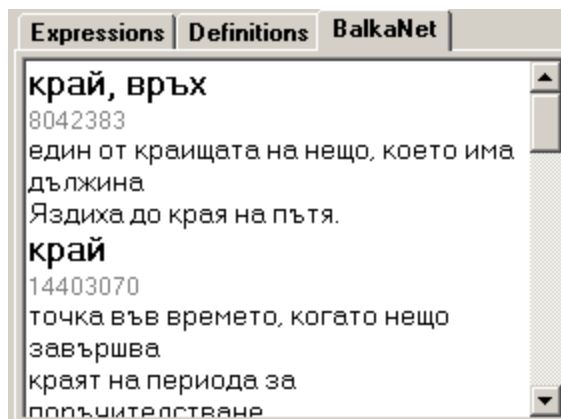


Fig 21. BalkaNet definitions

The second scenario is when the Wrapper is queried with a word, pointed in a context. This mode activates the most advanced component of the system - **the Word-Sense Disambiguator** - which definitely distinguishes “Smarty” from all the rest English-Bulgarian dictionaries and offers an advantage over the basic version of the comprehension assistant Xerox Locolex.

3.2.10. Word-Sense Disambiguator

The ultimate goal of a comprehension assistant is to provide only the most relevant translation of a word in a given context. To accomplish that, the first comprehension assistant – LocoLex – applies PoS-tagging and Multi-word expression recognition. “Smarty” also applies these techniques and then extends them with a more advanced matching.

3.2.10.1. Lexical Ambiguity

“Lexical ambiguity is a fundamental characteristic of language: words can have more than one distinct meaning. The 121 most frequent English nouns, which account for about one in five word occurrences in real text, have on average 7.8 meanings each”.[6] General verbs like “run”, “take” and “go” have even more than 40 senses each in WordNet 2.0.

3.2.10.2. Word-Sense Disambiguation

The process of identifying the right meaning of a homograph or a polysemous word, used in particular context, is called **word-sense disambiguation (WSD)**.

There are three main approaches for word-sense-disambiguation: [7, Ch.13]

1. Dictionary-based – introduced by Lesk in 1986. The method of Lesk and its modifications are based on the observation that the words in dictionary definitions could be used as “semantic tags” which represent connections between concepts.
2. Connectionists – they are based on research in psycholinguistics.
3. Statistical and machine learning – the rules for WSD are learned from semantic tagged examples and aligned bilingual corpora. The sets of senses emerge during the learning process.

3.2.10.3. Word-Sense Disambiguator in Smarty

Due to the limited time and resources, only a basic variation of the ideas of Lesk was implemented.

We use WordNet and BalkaNet as aligned semantic tagged dictionaries with glosses.

The process of WSD in “Smarty”:

1. A word in a text is pointed.
2. The word is tokenized and PoS-tagged together with the context.
3. The word and the context are sent to Word-Sense Disambigator.
4. Synsets from WordNet are found and glosses are extracted.
5. Context is lemmatized.

6. Lemmatized Context is cleaned from stop-words, considered to be confusing for the WSD: prepositions; “the”, “a”, “to”; pronouns, adverbs, conjunctions, numbers.
7. A cycle is started for each of the extracted synonym sets.
8. Each gloss is tokenized and part-of-speech tagged.
9. The gloss is lemmatized.
10. The lemmatized gloss is cleaned from irrelevant and stop-words.
(See 6)
11. The Context and the gloss are matched and number of word-matches is counted.
12. Until there are more glosses – go to 8. Else:
13. The gloss with higher number of matches is suggested, if there is one.
14. The synonym set of the suggested sense is matched to the synonym sets of BalkaNet.
15. If BalkaNet contains the disambiguated sense, then disambiguated translation in Bulgarian is displayed as well:

For as long as I can remember, I've been hearing that the world oil supply will run out in a few decades. I remember quite vividly growing up in the early 1980's hearing that there wouldn't be any oil left by the time I learned how to drive. Fortunately these predictions never came true, but the idea that we will exhaust all the oil under the **surface** of the earth is a persistent one.

8856787
the outermost level of the land or sea;
"earthquakes originate far below the
surface"; "three quarters of the Earth's
surface is covered by water"

земна повърхност

8856787
най-външната част от земната кора
Три четвърти от земната повърхност са покрити с
вода.

Fig 22. Word sense disambiguation windows

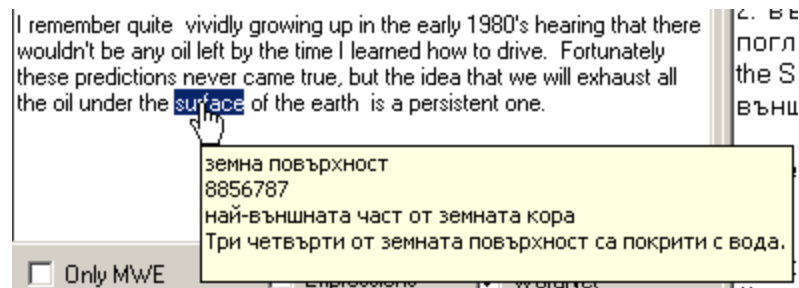


Fig 23. Disambiguated translation may appear as a tool tip, also.

Otherwise, if the synset exists in BalkaNet, but the precise sense is missing, then all senses are displayed with a question mark.

early 1980's hearing that there wouldn't be any oil left by the time I learned how to drive. Fortunately these predictions never came true, but the idea that we will exhaust all the oil under the surface of the earth is a persistent one.

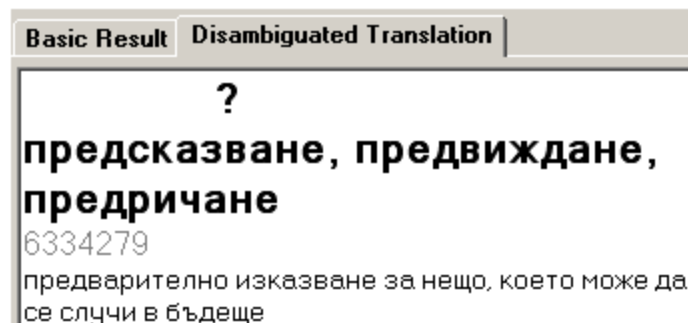


Fig 24. Uncertain disambiguated translation

In case the Disambiguator is unable to decide which one is the most appropriate sense, all senses from BalkaNet are displayed with a question mark, also.

3.3. Future Work

A number of possible improvements:

- More sophisticated multi-word expressions matching with improved scoring. In case of uncertainty, a list with suggestions should be presented to the user, instead of a single suggestion.
- More sophisticated implementation of Lesk's ideas for Word Sense Disambiguation and exploiting the semantic relations in WordNet and BalkaNet, which are not utilized yet.

Bulgarian Academy of Science is developing an extension of Bulgarian BalkaNet, called BulNet [3], which aims to increase the number of synsets with 15000, up to over 36000. This database will improve the performance of “Smarty” even without changes in Word Sense Disambiguation algorithm.

- Access to corpora and searching for passages and sentences in English, similar to the given in the input, via various matching techniques, semantic-distance evaluation and word sense disambiguation at paragraph-level.

Variations of HyperLex or/and PageRank graph algorithms, based on [1], might be tested. Although this would not be a translation, multiple examples may help reader to figure out patterns, thus - the meaning.

- Enriching multi-word expressions database.
- Development of Bulgarian-English direction in the system, intended for translation assistance.
- Web functions – Internet search for information or passages, which are relevant or similar to the context. Querying on-line dictionaries and combining results from many sources.

Several little tools are considered:

- Spell-checker – to correct typing mistakes when querying words in basic mode.
- Phonetic search – to display words that have similar transcriptions, not similar spelling.

4. Evaluation

4.1. Evaluation Method

Formal guidelines for comprehension assistant evaluation are not available. For that reason our evaluation will not be based upon a formal statistical comparisons.

“Smarty” is compared to other dictionaries and comprehension assistants, and both the advantageous aspects and its weaknesses are reviewed.

4.2. Interface Evaluation

4.2.1. Advantages

In comparison with the other English-Bulgarian dictionaries, “Smarty” has significantly more sophisticated and powerful user interface.

Translations on-click, multi-word expressions recognition and the popping-up tooltips impressed the few test users, who have seen or tried “Smarty”, during the development process. These features result in a significant speed-up in reading texts, which contain high number of unfamiliar words, because if using a dictionary with conventional interface, the user is supposed to switch manually between the window with the text and the window of the dictionary for every single word.

History list appears to be another useful element, when we come across a series of new words and we need a quick access to their translations, especially when working with texts on paper. Once entered, words could be quickly looked-up again without retyping or copying from do-it-yourself list from a text-editor.

4.2.1. Weaknesses

The graphical user interface could be more flexible – some windows could dynamically be hidden or made larger or smaller; font faces and sizes should be adjustable also.

History should contain part-of-speech and sense number/multi-word expression information, and the context where the word was found. Saving on disk should be available.

An option for tooltips, allowing them to appear without clicking, but immediately on pointing a word, also might be useful.

The standard dictionary entries output should be formatted better.

4.2.2. Smarty vs LocoLex

LocoLex laid the foundations of the architecture and user interface of the comprehension assistants, concentrating only on the most appropriate translation, presented in a minimalistic window. However, besides that, an option for browsing the conventional dictionary entries in the usual way also enriches the comprehension assistance experience, because some dictionary users enjoy reading all the meanings and exploring the content of the dictionaries. A similar reasoning applies for Suffix-search function, because it is intended to assist *browsing* for words with common endings.

4.3. Multi-Word Expressions Matching

As could be expected, better results are achieved in words-matching mode than in PoS-matching mode.

Shortest form of multi-word expressions, e.g. phrasal verbs like “run in”, “run out”, “take away” are almost always recognized correctly.

Longer expressions sometimes collide and the matcher returns multi-word

expression with a similar wording to the context, which is not the right one.

However, in the most cases, expressions are correctly recognized, and the results are adequate, even though the algorithm should be improved in future versions.

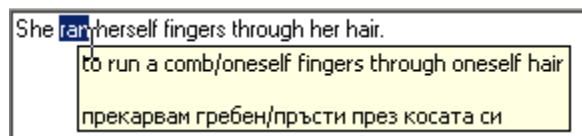


Fig 25. Multi-word expressions matching

4.4. Word Sense Disambiguation

Although the method for word sense disambiguation implemented in “Smarty” is simple (see 3.2.10.3), it achieves good results when appropriate words from the corresponding WordNet gloss are found in the context of the pointed word.

Unfortunately, the version of BalkaNet database that we use for providing disambiguated translations in Bulgarian consists of only 15000 synsets – compared to 115000 in WordNet 2.0, – and usually each of the synsets from BalkaNet has several times less senses than the corresponding synset in WordNet.

This is the reason why often correctly disambiguated English words could not be translated to Bulgarian at all, or if they can, the correct sense can not be found in the corresponding BalkaNet synset.

Nevertheless, the right gloss in English appears to be useful, as the user is expected to have a working knowledge of English.

4.4.1. Examples of correctly disambiguated and correctly translated senses

- *John works in a **bank**! He must have a lot of money...*

банка

7909067

финансова институция, която разполага с капитал и осъществява различни финансови операции
В тази банка направих ипотека на къщата.

bank - a financial institution that accepts deposits and channels the money into lending activities; "he cashed a check at the bank"; "that bank holds the mortgage on my home"

- *What instrument do you play, Paul?*

- *I play the **bass**.*

бас - музикален инструмент от група, който има най-ниско звучене

2330754

bass - the member with the lowest range of a family of musical instruments

- *You are **fired**!*

уволнявам, отстранявам от работа

2330754

въръчвам предизвестие за прекратяване на трудовия договор

2330754

fire - terminate the employment of; "The boss fired his secretary today"; "The company terminated 25% of its workers"

4.4.2. Example of correctly disambiguated word in English, without corresponding translation in BalkaNet:

*Mary walked to the **bank** and jumped in the river.*

8639924

Bank - sloping land (especially the slope beside a body of water);
"they pulled the canoe up on the bank"; "he sat on the bank of the river
and watched the currents"

?

банка

7909067

финансова институция, която разполага с капитал и осъществява
различни финансови операции
В тази банка направих ипотека на къщата.

рид, насип

8639793

продълговато хълмисто възвишение, образувано най-често от
ерозирането на външните земни пластове или на морското дъно
голям земен насип

5. Conclusion

This thesis has presented the framework for the next-generation intelligent multilingual computer dictionaries, called **comprehension assistants**, and the architecture of the newly developed English-Bulgarian comprehension assistant “**Smarty**”.

“Smarty” successfully applies the basic framework, which could be briefly described as: **Point a word in context**, **Part-of-speech tag** and **Multi-word expression recognize**. In addition, it extends the basic framework with **bilingual Word sense disambiguation**.

Although the most advanced feature was limited by the insufficiency of the available linguistic resources, “Smarty” demonstrates that the method is applicable and produces useful results.

The use of BulNet - a new WordNet for Bulgarian with a larger number of synonym sets – and the development of more sophisticated Word Sense Disambiguation algorithms that utilize the set of semantic relations in WordNet would further improve the results.

Finally, suggesting example sentences selected from corpora by applying semantic similarity measures and word sense disambiguation would provide even better comprehension assistance.

6. References

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...

Corrections: 20.7.2026

- * **The Sacred Computer:** Thinking Machines, Creativity and Human Development
<https://github.com/twenkid>
- * Thinking Machines 2026 / Self-Improving General Intelligence 2026: **SIGI-2026**
<https://github.com/twenkid/sigi-2026>